

Paper 20.50 - Layer-2 Synthesis Ledger Claim Contract

CQE-paper-20.50

CQECMPLX Formal Paper Series

Review-facing PDF built 2026-06-13

Construction Basis

This PDF is the clean paper artifact. Source extracts, kernels, receipts, tool notes, workbook material, and package bindings are used as construction evidence; they are not treated as separate review-facing papers.

Evidence Inputs

- top-level review source: Papers\Source\CQE-paper-20.50.md

Paper 20.50 - Layer-2 Synthesis Ledger Claim Contract

Purpose

This contract governs suite-level aggregation. It keeps solved, open, forbidden, and transported rows distinct.

Admitted Claims

Admit a row when it gives:

```
source paper
claim or route
classification
witness
proof boundary
current status
```

Rejected Promotions

Reject by default:

```
unknown -> solved
forbidden -> discarded
pass_with_open_lifts -> pass
registered target -> computed closure
accepted-looking row without validator
```

Hidden-Guess Rule

Diagnostics hide the answer until after the reviewer assigns:

```
solved
open
forbidden
transported
invalid
```

Linked Receipt

Current receipt:

```
production/formal-papers/CQE-paper-20/layer2_synthesis_ledger_receipt.json
```

Current verdict:

```
ledger accounting: pass
source-claim reproof by aggregation: not claimed
open lift closure: open
```

Boundary Failure

The contract fails if aggregation silently promotes rows.

It passes if aggregation preserves every row's source status.

Conclusion

Paper 20.50 keeps the synthesis layer honest.